

MAT 426, Class 6: Implementing GEPP

Two Important Coding Structures:

1. If/Then Statements:

- If a statement is _____, then _____.
- If a statement is _____, then we can execute a _____.
- More options can be implemented by using multiple logical statements and an if/else if structure
- Places we might see this:

- Syntax in Matlab/Octave:

2. For Loops

- Used to _____ a block of code a _____ number of times;
- Places we might see this:

- Syntax in Matlab/Octave:

Pieces of code we need to write:

- 1.
- 2.
- 3.
- 4.

Functions to Organize Code:

- 1.
- 2.
- 3.

LU FACTORIZATION USING GEPP

Input:

Outputs:

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For $k = 1, 2, \dots, n - 1$ do

1. (Find the pivot row).
2. (Interchange rows r_k and k).
3. (Form the multipliers).
4. (Update the entries).